Severn Lortie

EDUCATION

- **MASc in Robotics** · September 2025–Present
  Smith Engineering, Queen's University, Kingston
  Supervisor: Dr. Jonathan Gammell
  Thesis: *An Autonomous Sailboat and Informative Path Planner for Environmental Monitoring*
- **Bachelor of Computer Engineering** · September 2020–April 2025
  Smith Engineering, Queen's University, Kingston

RESEARCH EXPERIENCE

- **MASc Researcher** · September 2025–Present
  Robotic ESP Lab, Queen's University, Kingston
  Supervisor: Dr. Jonathan Gammell
  - Designed and deployed the complete control, actuation, and motion-planning stack for an autonomous sailboat
  - Validated the system through 6 on-water field tests in Lake Ontario
  - Achieved reliable waypoint following and optimal sail angle control, hitting 8/8 test waypoints in a summer 2026 field test
  - Designed and evaluated a sampling-based informative path planning algorithm that integrated sorted search heuristics for better performance over existing baselines
- **Field Assistant Technician - Drone Search and Rescue (SAR)** · September 2025–Present
  Robotic ESP Lab, Queen's University, Kingston
  Supervisor: Dr. Jonathan Gammell
  - Volunteered in support of drone SAR project at the lab. Key responsibilities include: piloting recovery boat for when drone landed, transporting and setting up equipment, and monitoring surrounding water and airspace for traffic.
  - Obtained pleasure craft and VHF radio license to support the safety of the operations.
- **Undergraduate Research Assistant** · May 2025–September 2025
  Robotic ESP Lab, Queen's University, Kingston
  - Programmed a software bridge to ingest data from NMEA 2000, a common protocol for marine sensors, into the Robot Operating System (ROS). This enabled the development of the autonomous sailboat for my Master's research. The code was open-sourced for other researchers to use

WORK EXPERIENCE

- **Software Engineering Intern** · May 2023–August 2024
  Distributive · Kingston, Ontario

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- Built and presented an LLM inference demo on Distributive's parallel-computing platform to Nokia's CTO and AI/HPC lead, contributing to follow-up meetings and a collaboration.
- Co-developed, from scratch, the company's Python API for its core parallel-computing platform in a two-engineer effort using an embedded JavaScript-in-Python runtime; achieved feature parity with the native JavaScript API and supported several non-trivial workloads.
- Designed a non-blocking WebSocket bridge in C++ from the embedded V8 workers to the Chrome DevTools Protocol; the resulting debugger was used to fix critical product bugs.

ENGINEERING DESIGN TEAMS AND PROJECTS

- **Ingenuity Labs Racing Team** · RoboRacer/F1Tenth · September 2026–Present
  Queen's University
  - Participated in three tests of the car and worked to diagnose complex fault, intermittent failure in the motor controller
  - Unified team workflow for track mapping and race line generation, allowing testing to be conducted without large setup overhead
  - Worked with team to diagnose fault that caused car to crash during previous competition, enabling more reliable operation
- **Reducing High Performance Computing Power Consumption with a Proactive CPU Governor** · Capstone Project · September 2024 · Queen's University
  - Worked with team of three other students to design a custom CPU frequency governor to increase the power efficiency of representative HPC workloads. Testing on a scientific proxy application called Quicksilver, the team was able to achieve a 4.6% decrease in runtime and 7.4% decrease in total energy consumption.
  - My role was to design and program the governor and create the final project poster.
- **Queen's VEX Robotics Team** · Firmware Team · September 2023–August 2024 · Queen's University
  - Devised high baud rate Lidar and Rotary Encoder communication channel using RP2040 microcontrollers for the team's robot.
  - The team passed qualifiers and competed using the communication subsystem at the annual VEX U Robotics World Championship in Dallas, Texas. The subsystem ran without issue during trials at competition, enabling the robot to perform its localization and mapping tasks.
- **Queen's Soft Robotics Team** · Electrical Team Lead · August 2022–April 2023 · Queen's University
  - Led a team researching lightweight, untethered power supplies for Peano-HASEL actuators.
  - Guided team members through creating circuit prototypes of high voltage step-up converters.
  - Played a key role in reaching a solution for a low-cost, low-power demand converter that could be used for the actuators.

PUBLICATIONS

1. Sanindie Silva, David Courtis, Jax Hodgkinson, and **Severn Lortie**. “Creating a Corn Yield Estimator: A Deep Learning Approach via UAV Imagery”
Canadian Undergraduate Conference on AI (CUCAI) · 2023

PRESENTATIONS AND POSTERS

- **Severn Lortie**, Meredith Scharlach, and Jonathan Gammell. “The Martin 16 Autonomous Sailboat”
Robotics and AI Symposium (RAIS) at Ingenuity Labs, Queen’s University
October 2026 · Poster
- Sanindie Silva, David Courtis, Jax Hodgkinson, and **Severn Lortie**. “Creating a Corn Yield Estimator: A Deep Learning Approach via UAV Imagery”
Canadian Undergraduate Conference on AI (CUCAI) · May 2023 · Oral presentation

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AWARDS AND SCHOLARSHIPS

- **Ontario Graduate Scholarship (OGS)** · $15,000 · 2026-2027
- **Vector Institute Scholarship in AI** · $17,500 · 2025-2026
- **NSERC Undergraduate Student Research Award (USRA)** · $9,000 · May 2025
- **Queen’s University Principal’s Scholarship** · $4,000 · 2020
- **2nd Place Paper** · Canadian Undergraduate Conference in AI (CUCAI) · $500 · May 2023
- **1st Place Project** · Amazon HACKTO Hackathon · $1,000 · September 2022

TEACHING EXPERIENCE

- **Teaching Assistant - Introduction to Computer Programming for Engineers** · Fall 2026
APSC 143 · Queen’s University
- **Teaching Assistant - Introduction to Computer Programming for Engineers** · Winter 2026
APSC 142 · Queen’s University
- **Teaching Assistant - Fundamentals of Information Structures** · Fall 2024 · ELEC 278
Queen’s University
- **Teaching Assistant - Data Structures and Algorithms** · Winter 2023 · MREN 178
Queen’s University

TECHNICAL SKILLS

- **Programming:** C/C++, Python, ROS, Julia, MATLAB/Simulink
- **Electrical design:** PCB design, low-level communication (CAN Bus, MODBUS, serial/RS422), LTSpice

- **Design and fabrication:** 3D printing, laser cutting, prototyping with CAD (OnShape and Fusion 360)

LEADERSHIP AND SERVICE

- **Lab Documentation Manager** · Robotic ESP Lab · September 2025–Present
  - Run the lab's Wiki and work with lab members to write high-quality documentation for their hardware and software designs
  - Manage public dataset hosting, both by setting up and maintaining server and also by working with university IT.